

TETRA-SHIELD AI

Contaminant Biotransformation Intelligence Report (offline cached data)

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TETRA-SHIELD PRIORITY SCORE is a computational prioritisation framework, not a validated clinical, regulatory, toxicological or environmental risk score. Predictions are labelled PREDICTED; literature-supported and experimentally demonstrated items are labelled as such.

CONTAMINANT — Tetracycline (C22H24N2O8, MW 444.4)

CANDIDATE — TetX2 monooxygenase [TETX2-BT] from *Bacteroides thetaiotaomicron*; evidence level E1

TETRA-SHIELD PRIORITY SCORE (BALANCED): 89.2 / 100 — rank #1

PRIORITISE for experimental validation | AMR caveat: enzyme is itself a resistance determinant — use only as cell-free purified biocatalyst, never as a released organism/gene.

Layer	Value 0–1	Layer	Value 0–1
evidence_strength	0.95	degradation_demonstration	0.95
structure_confidence	0.85	catalytic_plausibility	0.92
substrate_compatibility	0.95	transformation_confidence	0.85
circular_feasibility	0.62	environmental_relevance	0.80
product_safety_risk	0.30	residual_activity_risk	0.10
amr_risk	0.90	uncertainty_penalty	0.09
structural_uncertainty	0.15		

UNCERTAINTY FLAGS

product-tox-data-sparse

AMR SHIELD

HIGH CONCERN — enzymatic inactivation resistance gene (tetX2) (computational screening category, not a clinical/regulatory determination)

CORE DISTINCTIONS

REMOVAL != DEGRADATION | DEGRADATION != DETOXIFICATION | DOCKING != CATALYSIS | SEQUENCE SIMILARITY != FUNCTIONAL PROOF | PREDICTED STRUCTURE != EXPERIMENTAL STRUCTURE | NO AMR HIT != AMR-FREE | NO TOXICITY DATA != SAFE

SOURCES (candidate-scoped)

[L01] Yang W et al. 2004. TetX Is a Flavin-dependent Monooxygenase Conferring Resistance to Tetracycline Antibiotics. J Biol Chem 279(50):52346–52 — 10.1074/jbc.M409573200

[L02] Volkers G et al. 2011. Structural basis for a new tetracycline resistance mechanism relying on the TetX monooxygenase. J Mol Biol 416(5):2212–25 — 10.1016/j.jmb.2011.12.063

[L23] RCSB PDB: 2Y6R (TetX2 + 7-chlortetracycline + FAD, 3.1 Å), 4A6N (TetX + tigecycline, 2.3 Å), 1MNP (MnP, 2.0 Å), 1GYC (Trametes versicolor laccase, 1.9 Å) — 10.2210/pdb2Y6R/pdb